Chapter 27. The Volcano's Rim

The volcano-like surface that underlies much of this chapter (Figures 27.3, 27.5, and 27.9) can be described with elementary physics equations, as can the *Endurance*'s trajectory, the trajectory's instability on the rim, and the *Endurance*'s launch toward Miller's planet. See the appendix *Some Technical Notes*.

Chapter 28. Into Gargantua

In the Prologue of *Black Holes & Time Warps* (Thorne 1994), I describe, in much greater detail than here, what it would look like and feel like to fall through a black hole's horizon, both as seen and felt by the infalling person and as seen by someone else outside the black hole. And I describe how the look and feel are influenced by the mass of the black hole and by its spin.

Andrew Hamilton has constructed a "Black Hole Flight Simulator" for computing what it looks like to fall into a nonspinning black hole. His computations are similar to those done for *Interstellar* by Paul Franklin's team (Chapters 8, 9, and 15), but preceded *Interstellar* by many years. Andrew has used his simulator to produce a remarkable set of film clips that can be found on his website, http://jila.colorado.edu/~ajsh/insidebh, and in planetariums around the world (see http://www.spitzinc.com/fulldome_shows/show_blackholes).

Andrew's film clips differ from what we see in *Interstellar* in several ways: First, for pedagogical purposes Andrew sometimes paints a grid of lines on the black hole's horizon (there is no such grid for real black holes and none in *Interstellar*), and when he does so, he also replaces the star that imploded to form the black hole by a "past horizon."[59] Second, in his "Journey into a Realistic Black Hole," http://jila.colorado.edu/~ajsh/insidebh/realistic.html, Andrew endows